

Beyond Multimodal Gain: Sharp Partial Identification of Task-Relevant Latent-State Information

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Abstract

Predictive gains from an auxiliary modality are functions of the observed law. Claims that a report omits task-relevant information about an underlying state concern a latent functional. We formulate this distinction as a sharp partial-identification problem. Let $W = I(Y; M | T, C)$ denote the observable value of a proxy beyond a report and let $\Gamma = I(Y; E | T, C)$ denote the value of an admissible latent extension. At each report–context stratum, the proxy experiment is a Blackwell garbling of the latent experiment, so W lower-bounds Γ ; without further restrictions, the sharp identified set remains $[W, H(Y | T, C)]$. For channel classes with finite extreme-point representations on the refined side, a refinement–merging principle yields exact endpoint reductions for any continuous concave Bayes-risk functional. The lower endpoint is a finite-dimensional convex program, and the upper endpoint is a linear program when the relevant channel classes are polyhedral. Full-dimensional simplex classes give an explicit canonical coupling, observable compatibility conditions, and closed-form sharp bounds. Probability floors can force the sharp lower endpoint above the observed proxy value. Report-wise channel invariance alone does not couple latent states across report values; a full-support common-minorization restriction prevents perfect report-indexing and yields sharp cross-report bounds. We connect bounded latent cardinality to nonnegative factorization and develop cluster-aware inference for compatibility and identified endpoints. Simulations illustrate sharpness, coverage, and an identification–stability trade-off. In NHANES, objective accelerometer activity carries a robust positive mortality-relevant witness beyond self-reported activity; MELD and CrossCheck supply model-class and participant-clustered stress tests with weak or inconclusive evidence.

1 Introduction

Multimodal systems often improve prediction by adding audio, video, behavior, or physiology to a report. The gain is a property of observed variables. A scientific claim about information retained in the state that generated those variables concerns a different object.

Let E denote an unobserved state, T a report generated from that state, M a non-report proxy, Y a task outcome, and C observed context. The conditional mutual information

$$W := I(Y; M | T, C) \tag{1}$$

is identified from the observed law. It measures the value of one particular proxy after the report and context are known. The target

$$\Gamma := I(Y; E \mid T, C) \quad (2)$$

measures the task-relevant value of an admissible latent state beyond the report. A noisy proxy can carry a small fraction of the information present in that state. The observed law therefore identifies W and generally identifies only a set of possible values for Γ .

This distinction has a direct statistical-experiment interpretation. Conditional on a report–context stratum, the measurement model gives the Markov chain $Y - E - M$. The proxy experiment is a garbling of the latent experiment in the sense of Blackwell. Its value cannot exceed the value of the latent experiment for any decision problem. The ordering supplies a lower bound; the paper characterizes the sharp range of latent values compatible with the full observed law and explicit channel restrictions.

The target is task-specific. A report can be sufficient for one outcome and incomplete for another. It is also representation-specific in empirical work. If an analysis replaces T by a lossy representation $R = f(T)$, the measured latent gap includes the task-relevant report information discarded by R . We make this contribution explicit because it governs the interpretation of language embeddings, clusters, and model summaries.

The framework covers language-based human–AI interaction and a wider class of latent-measurement problems. Reports include symptom descriptions, survey responses, diagnostic codes, and self-reports. Proxies include sensors, behavior, second instruments, and modality-specific ratings. The design and assumptions determine a task-specific identified set for Γ ; its lower endpoint may be zero.

Contributions. Our contributions are threefold.

1. **A latent value-of-information estimand and its sharp benchmark.** We formulate task-relevant report insufficiency as partial identification, distinguish $I(Y; M \mid T, C)$ from $I(Y; E \mid T, C)$, and characterize the unrestricted sharp identified set. We also derive the exact change in the target induced by report coarsening.
2. **Exact endpoint geometry from refinement and merging.** For continuous concave Bayes-risk functionals, we show how opposite refinement and merging operations generate the two sharp endpoints. With a finite extreme-point representation on the refined channel, the lower endpoint is an exact finite-dimensional convex program. Polyhedral outcome and proxy classes yield a linear program for the upper endpoint. Full-dimensional simplex classes admit explicit canonical coordinates, observable compatibility conditions, and closed-form sharp bounds. Probability floors can make the lower endpoint strictly exceed the observable proxy value.
3. **Identification across reports through common minorization.** We show that report-wise channel invariance has no identifying force when latent states can carry a deterministic report index. A full-support common-minorization restriction prevents perfect report-indexing and induces cross-report coupling. The resulting finite-report model has sharp cross-report endpoints and observable tensor compatibility conditions.

We further connect bounded latent cardinality to nonnegative matrix and tensor rank, state the transition to classical point identification under tensor uniqueness, and give finite-sample procedures for compatibility and endpoint inference. Simulations study sharpness, coverage, and

decontamination stability. In NHANES, a finite-table analysis finds a robust positive accelerometer witness for future mortality beyond self-reported activity. Dialogue-grouped MELD and participant-clustered CrossCheck analyses provide model-class and longitudinal stress tests with weak or inconclusive evidence.

Organization. Section 2 positions the problem within statistical-experiment comparison, multimodal information decomposition, partial identification, and latent measurement. Section 3 defines the model, Blackwell ordering, statistical latent extensions, and the unrestricted benchmark. Section 4 develops the refinement–merging principle. Section 5 gives canonical simplex formulas and probability-floor bounds. Section 6 couples report strata through common minorization. Section 7 develops compatibility and endpoint inference. Section 8 reports simulations and the three empirical regimes. Extensions and proofs are collected in the appendices.

2 Statistical-Experiment View and Related Work

Comparison of experiments and value of information. Blackwell’s comparison of experiments orders signals by garbling and by performance across decision problems [Blackwell, 1951, 1953, Torgersen, 1991]. Le Cam’s treatment of sufficiency and approximate sufficiency extends experiment comparison beyond exact equivalence [Le Cam, 1964]. DeGroot measures information as the expected reduction in posterior uncertainty [DeGroot, 1962]; proper scoring rules and binary Bayes risks connect this view to concave entropies and divergences [Gneiting and Raftery, 2007, Reid and Williamson, 2011]. Data-processing inequalities describe the contraction of information along a fixed channel [Polyanskiy and Wu, 2017]. In our measurement model, the observed proxy is a garbling of the latent state within each report–context stratum. The proxy experiment and the complete observed law are fixed; the latent experiment ranges over extensions satisfying restrictions on the outcome, proxy, and report channels. The resulting object is the sharp range of the latent experiment’s value. Information-design models instead choose a signal structure to optimize an objective [Kamenica and Gentzkow, 2011].

Multimodal information decomposition. Multimodal learning studies how heterogeneous sources are represented, aligned, and combined [Baltrušaitis et al., 2019]. Partial information decomposition (PID) separates the information that observed sources carry uniquely, redundantly, or synergistically about a target [Williams and Beer, 2010, Bertschinger et al., 2014]. Alternative redundancy constructions use information geometry, pointwise surprisal, specificity and ambiguity lattices, and source orderings [Harder et al., 2013, Ince, 2017, Finn and Lizier, 2018, Kolchinsky, 2022]. Kolchinsky’s Blackwell-order construction defines redundancy among observed sources [Kolchinsky, 2022]. Multimodal applications estimate interactions among observed modalities or model-induced predictive distributions [Liang et al., 2023, Fang et al., 2026]. Here the complete observed law is fixed, the optimization ranges over admissible latent extensions, and the output is the sharp range of $I(Y; E \mid T, C)$. Table 1 summarizes the two formulations.

Partial identification with latent variables and proxy measurements. Partial-identification theory characterizes the values compatible with the observed distribution and maintained assumptions [Manski, 2003, Tamer, 2010, Beresteanu et al., 2011, Kline and Tamer, 2023, Molinari, 2020]. Contamination, measurement error, and misclassification motivate bounds and set identification when latent quantities are observed imperfectly [Horowitz and Manski, 1995, Molinari, 2008, Schennach, 2016, Hu, 2008]. Recent work extends this perspective to machine-generated measurements. Chen

and Tamer [2026] study latent prevalence from panels of LLM reports under externally calibrated restrictions. Li [2026] study downstream moment parameters linked to machine-learned proxies and validation data, and Ogut and Yin [2026] analyze a latent regressor measured through multiple nonlinear proxies. Our estimand is a conditional value-of-information functional. Its identified set follows from restrictions on the outcome, proxy, and report channels and from observable compatibility with those restrictions.

Table 1: Closest-neighbor comparison with multimodal PID. The mathematical distinction is determined by the variables, the object held fixed, the optimization domain, and the output.

	Multimodal PID	This paper
Variables	Observed modalities and a target; often model-induced predictive distributions	Observed (T, M, Y, C) and latent state E
Held fixed	Observed or model distribution and selected marginals	Complete observed law of (T, M, Y, C)
Optimization domain	Couplings or decompositions defining unique, redundant, and synergistic information	Latent extensions and factorizations satisfying channel restrictions
Primary target	Allocation of task-relevant information among observed sources	$I(Y; E \mid T, C)$
Output	Information decomposition	Sharp identified interval, compatibility region, and endpoint inference
Question	How do observed modalities share and combine predictive information?	What latent value of information is compatible with the observed proxy and report?

Latent-class and multi-view identification. Latent-class and multi-view models combine conditional independence with rank, separability, or genericity conditions to identify mixture components [Hui and Walter, 1980, Hall and Zhou, 2003, Kruskal, 1977, Allman et al., 2009, Anandkumar et al., 2014]. Related work on nonnegative matrix factorization gives conditions for full factor identifiability [Fu et al., 2018]. Gillis and Rajkó [2023] study partial identifiability as uniqueness of selected factor columns. Our target remains well defined when the latent decomposition is nonunique: it is a functional evaluated over all admissible decompositions. Nonnegative rank controls feasibility and endpoint attainment, and tensor uniqueness marks the point-identification boundary.

Decontamination, common information, and report overlap. Mutual-contamination models recover component distributions from observed convex mixtures [Katz-Samuels et al., 2019]. The simplex transformation used here is algebraically related to demixing and produces a canonical extreme-coordinate coupling for compatibility and endpoint calculations. Gács–Körner and Wyner common information define distinct forms of information shared by observed variables [Gács and Körner, 1973, Wyner, 1975]; PID constructions optimize over other auxiliary variables or couplings [Bertschinger et al., 2014]. Common minorization serves a separate role here. Doebelin coefficients quantify shared mass and contraction in Markov kernels [Makur and Singh, 2024]; fixing a full-support minorization level restricts the report channel and rules out report-specific state duplication.

Inference for identified sets and compatibility. Inference for partially identified parameters distinguishes coverage of a scalar value, a projection, and the full identified set [Imbens and

Manski, 2004, Chernozhukov et al., 2007, Romano and Shaikh, 2010, Kaido et al., 2019]. Our compatibility restrictions form a finite collection of moment inequalities. Generalized moment selection and Gaussian multiplier approximations provide tools for simultaneous inference on such collections [Andrews and Soares, 2010, Chernozhukov et al., 2013]. Cluster-level resampling addresses dependence within subjects, dialogues, or sampling units [Cameron et al., 2008]. We specialize these ideas to canonical compatibility cells and project uncertainty in the observed law through the sharp endpoint maps.

3 Setup, Experiment Ordering, and the Identification Benchmark

All logarithms are base two. Random variables are defined on a common probability space. The main results condition on $C = c$ and suppress c when no ambiguity arises.

3.1 Observed law and latent extensions

Let

$$O = (Y, M, T, C)$$

denote the observed data. The variables have the following roles.

- E is an unobserved state relevant to the task.
- T is a report or report-derived observation.
- M is a non-report proxy for the state.
- Y is the task outcome.
- C contains observed context.

The core measurement model is

$$p(y, m, t \mid e, c) = q_{e,c}(y) r_{e,c}(m) s_{e,c}(t). \quad (3)$$

Thus Y , M , and T are conditionally independent views of E after conditioning on C .

Definition 3.1 (Statistical latent extension and identified set). *A statistical latent extension of an observed law P_O is a joint law P on (E, Y, M, T, C) whose observed marginal is P_O and which satisfies (3). For a stated extension class $\mathcal{M}(P_O)$, define*

$$W(P_O) := I(Y; M \mid T, C), \quad (4)$$

$$\Gamma(P) := I_P(Y; E \mid T, C), \quad (5)$$

$$\mathcal{I}_\Gamma(P_O) := \{\Gamma(P) : P \in \mathcal{M}(P_O)\}. \quad (6)$$

The theorems are relative to the declared extension class. A scientific application may restrict that class through temporal ordering, nonanticipation, exclusion restrictions, or external measurement validation. Such restrictions can narrow the identified set. The unrestricted endpoint constructions below are statistical saturation devices. A scientifically prior-state interpretation requires the corresponding temporal and causal restrictions.

The target is tied to Y . State variation unrelated to the task does not contribute to Γ .

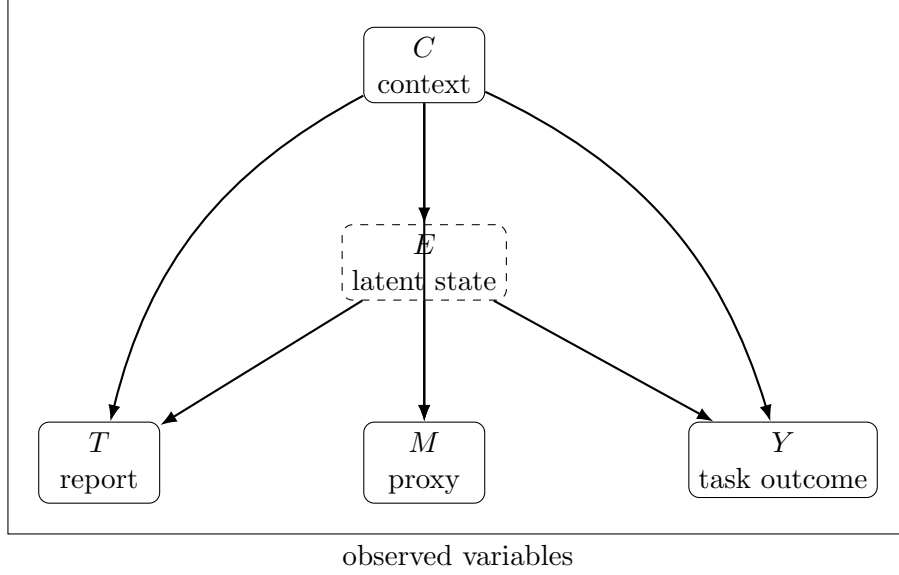


Figure 1: Measurement structure. The identified target is defined over latent extensions that reproduce the observed law and satisfy the conditional factorization.

3.2 Value of information and Blackwell order

Fix $X = (T, C) = x$. For a signal S , let $p_{Y|S,x}$ denote the posterior distribution of Y and let $\Phi : \Delta_L \rightarrow \mathbb{R}$ be a continuous concave Bayes-risk functional. Define the value of the signal by

$$V_\Phi(S; x) := \Phi(p_{Y|x}) - \mathbb{E}[\Phi(p_{Y|S,x}) | X = x]. \quad (7)$$

This is DeGroot's expected reduction in uncertainty for a general uncertainty function [DeGroot, 1962]. Proper scoring losses generate such concave generalized entropies [Gneiting and Raftery, 2007]. Shannon entropy gives

$$V_H(S; x) = I(Y; S | X = x).$$

Proposition 3.2 (Blackwell ordering of the latent and proxy experiments). *Under (3), the experiment generated by M about state Y is a Blackwell garbling of the experiment generated by E , conditional on $X = x$. Hence*

$$\boxed{V_\Phi(E; x) \geq V_\Phi(M; x)} \quad (8)$$

for every continuous concave Bayes-risk functional Φ . For $\Phi = H$,

$$I(Y; E | X = x) \geq I(Y; M | X = x). \quad (9)$$

Indeed, $p_{Y|M,x} = \mathbb{E}[p_{Y|E,x} | M, x]$. Jensen's inequality gives (8). This is the posterior form of Blackwell monotonicity [Blackwell, 1951, 1953, Torgersen, 1991]. The rest of the paper asks how much stronger the inequality must be once the latent experiment and its garbling channels are restricted.

Under logarithmic loss,

$$H(Y | T, C) - H(Y | E, C) = I(Y; E | T, C) = \Gamma. \quad (10)$$

The acquisition target can be separated from representation and prediction error. If a system maps (T, C) to Z and outputs $q_\theta(Y | Z)$, then

$$\mathbb{E}[-\log q_\theta(Y | Z)] - H(Y | E, C) = I(Y; E | T, C) + I(Y; T, C | Z) + \mathbb{E}_Z D_{\text{KL}}(p(Y | Z) \| q_\theta(Y | Z)). \quad (11)$$

The identification results concern the first term.

3.3 Unrestricted benchmark

A binary example makes the identification problem explicit. Let T and C be constant, let Y be a fair bit, and let $M = Y \oplus N$ with $N \sim \text{Bernoulli}(0.2)$. The observed witness is $1 - h_2(0.2)$. The extension $E = M$ gives the same value for Γ , while $E = (M, Y)$ gives $\Gamma = 1$. Both reproduce the same observed table.

Proposition 3.3 (Unrestricted sharp benchmark). *Let Y be finite. Over all statistical latent extensions satisfying (3), with no channel or cardinality restrictions,*

$$\boxed{\mathcal{I}_\Gamma(P_O) = [I(Y; M | T, C), H(Y | T, C)]}. \quad (12)$$

The lower endpoint follows from

$$\Gamma = I(Y; M | T, C) + I(Y; E | T, M, C), \quad (13)$$

which holds under $Y \perp\!\!\!\perp M | E, T, C$. It is attained by $E = (T, M)$. The upper endpoint is attained by $E = (T, M, Y)$. Mixing the two extensions fills the interval. [Section B.1](#) gives the proof.

The benchmark isolates the source of nonidentification. Informative bounds require restrictions that exclude deterministic proxy and outcome readouts or limit report-specific refinement.

3.4 Report representations

Empirical analyses commonly replace T by a representation $R = f(T)$. This operation changes the estimand.

Lemma 3.4 (Representation coarsening). *Let $R = f(T)$ and assume $Y \perp\!\!\!\perp T | E, C$. Then*

$$\boxed{I(Y; E | R, C) = I(Y; E | T, C) + I(Y; T | R, C)}. \quad (14)$$

The second term is report information lost by the representation. An empirical gap based on an embedding or cluster is therefore a representation-relative target Γ_R . This identity separates report insufficiency from information discarded by the analysis pipeline.

Four quantities must remain distinct in applications. A fitted predictor can estimate a model-class score gain without consistently estimating the observable-law information functional; the latter is still only a witness for a statistical latent extension; and a scientific prior-state interpretation requires additional temporal and measurement structure.

Table 2: Empirical and latent quantities used in the paper. Equality between rows requires additional conditions and is never assumed by notation alone.

Object	Definition	What identifies or estimates it	Interpretive boundary
Model score gain	Difference in out-of-sample loss within fitted model classes	Cross-fitted predictions and a declared scoring rule	Can be negative and need not equal mutual information
Observable witness	$W = I(Y; M \mid T, C)$	The observed joint law; a finite-table plug-in targets it directly	Information in one measured proxy, not all information in E
Statistical latent value	$\Gamma = I(Y; E \mid T, C)$	An identified set over admissible latent extensions	Endpoint constructions may be statistical saturation devices
Scientific prior-state value	$I(Y; E_{\text{pre}} \mid T, C)$	The observed law plus temporal, exclusion, and measurement-validity restrictions	Requires a substantively justified definition of E_{pre}

3.5 Structural restrictions and their roles

Table 3 summarizes the restrictions used below. Compatibility with the observed law is an empirical condition. Calibration supplies the substantive content of the numerical parameter values.

Table 3: Structural objects, identification roles, and possible calibration sources.

Object	Restriction	Identification role	Possible basis
Measurement structure	Y, M, T conditionally independent given (E, C)	Makes report and proxy measurements of a common latent extension	Temporal design, independent measurement protocols, negative controls
Outcome channel	$q_e(y) \geq a_y$	Limits how completely E can determine Y ; lowers the upper endpoint	Repeated outcomes, domain bounds, validation data
Proxy channel	$r_e(m) \geq b_m$	Excludes a noiseless proxy; raises the lower endpoint	Repeated sensors or raters, error studies, validation sample
Report channel	$s_e \geq \delta\nu$, with full-support ν	Prevents perfect report-indexing and induces cross-report coupling	Repeated elicitation, experimental design, fixed reference distribution
Latent complexity	$ \mathcal{E} \leq K$	Limits state refinement and introduces factorization constraints	Domain model, design restriction, parsimonious state representation
Report representation	$R = f(T)$	Changes the target by adding discarded report information	Encoder comparison, sufficiency checks, representation audit
Proxy representation	$M = g(M_{\text{raw}})$	Changes both W and the numerical meaning of proxy floors	Frozen binning, sensor validation, representation ladder

4 Sharp Endpoints by Refinement and Merging within Report Strata

Fix

$$X = x, \quad X := (T, C),$$

and let P_x be the $L \times J$ table of (Y, M) in that stratum. The observed proxy experiment is a garbling of an unknown latent experiment. We restrict the outcome posteriors and proxy garbling channels and optimize the latent value of information over factorizations

$$P_x = \sum_{e=1}^K \pi_e q_e r_e^\top. \quad (15)$$

Here $q_e \in \Delta_L$ is the posterior law of Y given the latent state and $r_e \in \Delta_J$ is the proxy channel.

Let $\Phi : \Delta_L \rightarrow \mathbb{R}$ be continuous and concave. The local value of the latent experiment is

$$\Gamma_{\Phi, x} := \Phi(p_{Y|x}) - \sum_e \pi_e \Phi(q_e). \quad (16)$$

This is the Jensen gap generated by the posterior split $\{(\pi_e, q_e)\}$. For $\Phi = H$, it equals $I(Y; E | X = x)$. More generally,

$$\Phi_\ell(q) = \inf_a \sum_y q_y \ell(y, a)$$

is concave for every decision loss ℓ .

For $z \in \mathbb{R}_+^L$, define the concave perspective

$$\tilde{\Phi}(z) := \begin{cases} (\mathbf{1}^\top z) \Phi(z/(\mathbf{1}^\top z)), & \mathbf{1}^\top z > 0, \\ 0, & z = 0. \end{cases} \quad (17)$$

For a convex set $\mathcal{Q} \subseteq \Delta_L$, write

$$\text{cone}(\mathcal{Q}) = \{\lambda q : \lambda \geq 0, q \in \mathcal{Q}\}.$$

4.1 The refinement–merging principle

Theorem 4.1 (Lower endpoint by proxy refinement). *Let $\mathcal{Q} \subseteq \Delta_L$ be compact and convex, and let*

$$\mathcal{R} = \text{conv}\{r^1, \dots, r^B\} \subseteq \Delta_J.$$

Suppose the feasible set is nonempty. If $K \geq B$, the sharp lower endpoint of (16) over decompositions (15) with $q_e \in \mathcal{Q}$ and $r_e \in \mathcal{R}$ is

$$\boxed{\Gamma_{\Phi, x} = \Phi(p_{Y|x}) - \max_{z_1, \dots, z_B} \sum_{j=1}^B \tilde{\Phi}(z_j)} \quad (18)$$

subject to

$$P_x = \sum_{j=1}^B z_j (r^j)^\top, \quad z_j \in \text{cone}(\mathcal{Q}). \quad (19)$$

When the extreme points r^j are available, this is an exact finite-dimensional convex program.

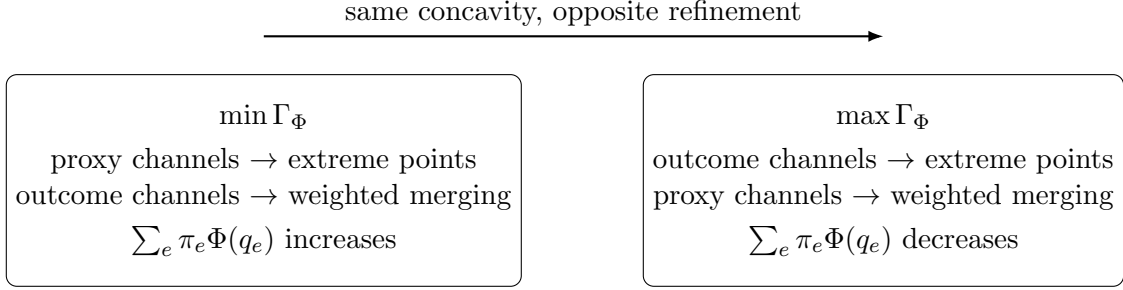


Figure 2: The refinement–merging principle. The endpoint determines which channel is refined and which latent components are merged.

The construction refines each proxy channel into extreme proxy channels and merges latent states that share the same proxy coordinate. Merging replaces their outcome posteriors by a weighted average. Concavity raises the conditional Bayes risk and lowers the value of the latent experiment.

Theorem 4.2 (Upper endpoint by outcome refinement). *Let*

$$\mathcal{Q} = \text{conv}\{u^1, \dots, u^A\} \subseteq \Delta_L$$

and let $\mathcal{R} \subseteq \Delta_J$ be compact and convex. Suppose the feasible set is nonempty. If $K \geq A$, the sharp upper endpoint is

$$\bar{\Gamma}_{\Phi, x} = \Phi(p_{Y|x}) - \min_{m_1, \dots, m_A} \sum_{i=1}^A (\mathbf{1}^\top m_i) \Phi(u^i) \quad (20)$$

subject to

$$P_x = \sum_{i=1}^A u^i m_i^\top, \quad m_i \in \text{cone}(\mathcal{R}). \quad (21)$$

If $\mathcal{R}$ is polyhedral, the endpoint is the value of a linear program after homogenizing the membership constraints for m_i .

The upper construction refines outcome posteriors to extreme points. Concavity lowers their average Bayes risk and raises the value of the latent experiment. States with the same outcome extreme point can be merged on the proxy side.

The two theorems sharpen the Blackwell lower bound by exploiting the geometry of admissible latent experiments. The observable law fixes the proxy experiment; the endpoint programs select the least and most valuable latent experiments that can generate it through allowed channels.

Remark 4.3 (Cardinality). *The conditions $K \geq B$ and $K \geq A$ are part of the endpoint statements. The lower construction uses at most one state per proxy vertex, and the upper construction uses at most one state per outcome vertex after merging. A binding cardinality limit recouples the channels through nonnegative factorization. [Section A.1](#) gives the exact rank formulation for simplex models.*

Remark 4.4 (Computation). *The finite convex formulation assumes that the required extreme points are available. An H -polytope can have exponentially many vertices. The probability-floor simplexes in [Section 5](#) have exactly L and J vertices and lead to explicit formulas.*

5 Simplex Channels: Canonical Coordinates and Closed-Form Bounds

Full-dimensional simplex channel classes turn the endpoint reductions into linear coordinate changes. They include asymmetric probability floors and yield observable compatibility conditions together with explicit sharp endpoints.

5.1 Canonical coupling in simplex coordinates

Let

$$\mathcal{Q} = \text{conv}\{u^1, \dots, u^L\}, \quad \mathcal{R} = \text{conv}\{v^1, \dots, v^J\},$$

where the vertex matrices

$$U = [u^1 \ \dots \ u^L], \quad V = [v^1 \ \dots \ v^J]$$

are invertible. Every admissible channel has unique simplex coordinates:

$$q_e = Ua_e, \quad r_e = Vb_e, \quad a_e \in \Delta_L, \ b_e \in \Delta_J.$$

Equation (15) becomes

$$P_x = U\Omega_x V^\top, \quad \Omega_x := \sum_e \pi_e a_e b_e^\top. \quad (22)$$

Thus

$$\boxed{\Omega_x = U^{-1} P_x V^{-\top}.} \quad (23)$$

The matrix Ω_x is the coupling induced on the extreme coordinates of the two channel simplexes. The observed table and the chosen channel classes identify this canonical object. Different latent-state factorizations can induce the same Ω_x .

Theorem 5.1 (Canonical simplex representation and sharp endpoints). *With unrestricted latent cardinality, the observed table P_x is compatible with the simplex channel model if and only if*

$$\Omega_x \geq 0 \quad (24)$$

entrywise. When this condition holds, define

$$z_j := P_x V^{-\top} e_j, \quad \lambda_j := \mathbf{1}^\top z_j, \quad \bar{q}_j := z_j / \lambda_j, \quad (25)$$

$$m_i^\top := e_i^\top U^{-1} P_x, \quad \gamma_i := \mathbf{1}^\top m_i. \quad (26)$$

Zero-mass components are omitted. For Shannon entropy,

$$\boxed{\underline{\Gamma}_x = H(p_{Y|x}) - \sum_{j=1}^J \lambda_j H(\bar{q}_j)} \quad (27)$$

$$\boxed{\bar{\Gamma}_x = H(p_{Y|x}) - \sum_{i=1}^L \gamma_i H(u^i)} \quad (28)$$

Both endpoints are sharp when the corresponding cardinality conditions in [Theorems 4.1](#) and [4.2](#) hold.

The compatibility statement follows because any latent model induces a nonnegative coupling in (22). Conversely, a nonnegative Ω_x gives a cell construction with latent state (i, j) and weight $(\Omega_x)_{ij}$.

5.2 Asymmetric probability floors

Let $a \in \mathbb{R}_+^L$ and $b \in \mathbb{R}_+^J$ satisfy

$$A := \mathbf{1}^\top a < 1, \quad B := \mathbf{1}^\top b < 1.$$

Consider

$$\mathcal{Q}_a := \{q \in \Delta_L : q_y \geq a_y \ \forall y\}, \quad (29)$$

$$\mathcal{R}_b := \{r \in \Delta_J : r_m \geq b_m \ \forall m\}. \quad (30)$$

The vertices are

$$u^i = a + (1 - A)e_i, \quad v^j = b + (1 - B)e_j,$$

with

$$U = (1 - A)I + a\mathbf{1}^\top, \quad U^{-1} = \frac{I - a\mathbf{1}^\top}{1 - A}, \quad (31)$$

$$V = (1 - B)I + b\mathbf{1}^\top, \quad V^{-1} = \frac{I - b\mathbf{1}^\top}{1 - B}. \quad (32)$$

Write $p_{ym} = P(Y = y, M = m \mid X = x)$, with marginals p_y and p_m . The canonical coupling has entries

$$(\Omega_x)_{ym} = \frac{p_{ym} - a_y p_m - b_m p_y + a_y b_m}{(1 - A)(1 - B)}. \quad (33)$$

Hence the model is compatible exactly when the numerator in (33) is nonnegative for every (y, m) .

Define

$$\lambda_m := \frac{p_m - b_m}{1 - B}, \quad (34)$$

$$q_y^{(m)} := \frac{p_{ym} - b_m p_y}{p_m - b_m}, \quad (35)$$

$$\gamma_y := \frac{p_y - a_y}{1 - A}. \quad (36)$$

The canonical $q^{(m)}$ is well defined whenever $\lambda_m > 0$.

Corollary 5.2 (Closed-form probability-floor interval). *If (33) is nonnegative entrywise, then*

$$\underline{\Gamma}_x = H(p_Y) - \sum_m \lambda_m H(q^{(m)}) \quad (37)$$

$$\bar{\Gamma}_x = H(p_Y) - \sum_y \gamma_y H(a + (1 - A)e_y) \quad (38)$$

The lower endpoint is controlled directly by the proxy floor b , while the upper endpoint is controlled directly by the outcome floor a . The two restrictions interact through the compatibility inequalities.

The symmetric case sets $a_y = \alpha$ and $b_m = \eta$. Then

$$\bar{\Gamma}_x = H(p_Y) - h_L(\alpha), \quad (39)$$

where

$$h_L(\alpha) := -[1 - (L - 1)\alpha] \log_2[1 - (L - 1)\alpha] - (L - 1)\alpha \log_2 \alpha. \quad (40)$$

Positive a_y imposes residual outcome randomness within every latent state. Positive b_m excludes a deterministic proxy readout. Their scientific calibration is separate from observable compatibility; Table 3 lists possible sources.

5.3 Strict improvement over the observable witness

The lower construction in (37) introduces a canonical proxy coordinate $J^\star$ with distribution λ and conditional outcome law $q^{(m)}$. The channel $J^\star \rightarrow M$ has columns v^m . It satisfies

$$Y - J^\star - M$$

within the stratum $X = x$, and

$$\underline{\Gamma}_x = I(Y; J^\star \mid X = x).$$

Lemma 5.3 (Strict witness improvement). *Assume $b_m > 0$ for every m , the probability-floor model is compatible, and*

$$I(Y; M \mid X = x) > 0.$$

Then

$$\boxed{\underline{\Gamma}_x > I(Y; M \mid X = x)}. \quad (41)$$

The result follows from strict data processing for the full-support channel $J^\star \rightarrow M$ [Polyanskiy and Wu, 2017]. Equality would require M to remain sufficient for the Y -relevant distinctions in $J^\star$. Full support connects every active input state to every output state; the equality condition then forces the conditional law of Y to be constant across active inputs, contradicting positive information. Section B.3 gives the argument.

Example 5.4 (Binary illustration). *Let $Y, M \in \{0, 1\}$, impose symmetric floors α and η , and observe the 2×2 table P_x . The compatibility constraints in (33) are four linear inequalities. At $\eta = 0$, the lower endpoint equals $I(Y; M \mid x)$. Any compatible $\eta > 0$ raises the lower endpoint when the witness is positive. At $\alpha = 0$, the upper endpoint is $H(Y \mid x)$; positive α lowers it by the minimum entropy retained inside every latent state.*

6 Coupling Report Strata through Common Minorization

The stratum-wise analysis allows a separate latent decomposition for every report value. A shared channel notation does not force those decompositions to share latent states.

6.1 Why channel invariance alone is insufficient

Proposition 6.1 (No free tightening). *Suppose that, for each report-context stratum (t, c) ,*

$$P(Y, M \mid t, c) = \sum_h \pi_{h|t,c} q_{h,t,c} r_{h,t,c}^\top. \quad (42)$$

If latent cardinality is unrestricted and the report channel may be deterministic, the collection of stratum-specific models can be represented by a single latent variable satisfying (3). One construction sets $E' = (t, h)$ and makes T deterministic given E' .

The proposition isolates the missing ingredient: cross-report coupling. Report-wise invariance has identifying content only when the model prevents latent states from carrying the report label exactly. We impose a common-minorization restriction on the report channel. It prevents perfect report-indexing; it does not require different report values to share an otherwise identical scientific state.

6.2 Finite report alphabets

Fix $C = c$. Let Y , M , and T have alphabet sizes L , J , and G . The observed law is

$$\mathcal{P} = [p_{ymt}] = \sum_{e=1}^K \pi_e q_e \otimes r_e \otimes s_e. \quad (43)$$

We impose

$$q_e \in \mathcal{Q}_a := \{q \in \Delta_L : q \geq a\}, \quad A := \mathbf{1}^\top a < 1, \quad (44)$$

$$r_e \in \mathcal{R}_b := \{r \in \Delta_J : r \geq b\}, \quad B := \mathbf{1}^\top b < 1, \quad (45)$$

$$s_e \in \mathcal{S}_d := \{s \in \Delta_G : s \geq d\}, \quad \delta := \mathbf{1}^\top d < 1. \quad (46)$$

When $\delta > 0$, write $d = \delta\nu$ for a reference distribution $\nu \in \Delta_G$. Then

$$s_e = \delta\nu + (1 - \delta)\tilde{s}_e, \quad (47)$$

so every latent-state report distribution contains the same component ν with mass δ . A full-support ν rules out deterministic report labels; a restricted-support ν couples the strata in its support. The maximal common mass, optimized over ν , is the Doeblin coefficient of the report kernel. In the identification analysis, (δ, ν) is fixed by calibration or sensitivity analysis.

The simplex vertex matrices are

$$U = (1 - A)I + a\mathbf{1}^\top, \quad V = (1 - B)I + b\mathbf{1}^\top, \quad S = (1 - \delta)I + d\mathbf{1}^\top. \quad (48)$$

Define the canonical tensor

$$\boxed{\mathcal{W} := \mathcal{P} \times_1 U^{-1} \times_2 V^{-1} \times_3 S^{-1}.} \quad (49)$$

The notation $\times_k$ denotes mode- k multiplication.

Theorem 6.2 (Cross-report canonical representation and sharp endpoints). *With unrestricted latent cardinality, the model in (43)–(46) is compatible with $\mathcal{P}$ if and only if*

$$\mathcal{W} \geq 0 \quad (50)$$

entrywise. When compatible, define

$$\mathcal{Z} := \mathcal{P} \times_2 V^{-1} \times_3 S^{-1}, \quad (51)$$

with slices $z^{jk} := \mathcal{Z}_{\cdot jk}$,

$$\lambda_{jk} := \mathbf{1}^\top z^{jk}, \quad q^{jk} := z^{jk} / \lambda_{jk}.$$

Then the sharp lower endpoint of

$$\Gamma_c = I(Y; E \mid T, C = c)$$

is

$$\boxed{\underline{\Gamma}_c = H(Y \mid T, c) - \sum_{j,k} \lambda_{jk} H(q^{jk}).} \quad (52)$$

Let $u^i = a + (1 - A)e_i$ and

$$\gamma_i = \frac{P(Y = i \mid c) - a_i}{1 - A}. \quad (53)$$

The sharp upper endpoint is

$$\bar{\Gamma}_c = H(Y \mid T, c) - \sum_i \gamma_i H(u^i). \quad (54)$$

The lower endpoint is attained with at most

$$N_- := \#\{(j, k) : \lambda_{jk} > 0\}$$

latent states.

The compatibility condition is observable. Expanding (49) gives

$$\begin{aligned} (1 - A)(1 - B)(1 - \delta)\mathcal{W}_{ijk} &= p_{ijk} - a_i p_{\cdot jk} - b_j p_{i \cdot k} - d_k p_{ij \cdot} \\ &\quad + a_i b_j p_{\cdot \cdot k} + a_i d_k p_{\cdot j \cdot} + b_j d_k p_{i \cdot \cdot} - a_i b_j d_k. \end{aligned} \quad (55)$$

The lower construction refines the proxy and report channels and merges states with the same extreme-coordinate pair (j, k) . The upper construction refines the outcome channel. The same refinement–merging principle therefore governs both the stratum-wise and cross-report models.

Corollary 6.3 (Stratified limit). *If $d = 0$, then $S = I$ and*

$$\underline{\Gamma}_c = \sum_t P(t \mid c) \underline{\Gamma}_{t,c}^{\text{strat}}, \quad (56)$$

where $\underline{\Gamma}_{t,c}^{\text{strat}}$ is the lower endpoint from [Section 4](#) for $P(Y, M \mid t, c)$.

If $\mathcal{S}_{d_2} \subseteq \mathcal{S}_{d_1}$, then the admissible extension class under d_2 is smaller and

$$\underline{\Gamma}(d_2) \geq \underline{\Gamma}(d_1). \quad (57)$$

Thus common report mass tightens partial identification by limiting perfect report-indexing across strata. States may remain nearly report-specific when δ is small.

6.3 Boundary regimes and extensions

A binding state-count restriction recouples the channel classes. In the finite-view simplex model, the minimum compatible latent cardinality is the nonnegative CP rank of the canonical tensor,

$$\min |\mathcal{E}| = \text{rank}_+^{\text{CP}}(\mathcal{W}). \quad (58)$$

The minimum cardinality required to attain the upper endpoint is a sum of nonnegative matrix ranks over the outcome-coordinate slices. [Section A.1](#) gives the exact statements and proofs.

Classical tensor-uniqueness conditions define the point-identification boundary [[Kruskal, 1977](#), [Allman et al., 2009](#), [Anandkumar et al., 2014](#)]. If the three factor matrices satisfy the usual Kruskal rank condition, the latent decomposition is unique up to label permutation and Γ_c is point identified. Common minorization and rank diversity have separate roles: the first narrows the extension class; the second can make the decomposition unique.

The population result also extends beyond finite report alphabets. [Section A.2](#) treats T on a standard Borel space with a fixed reference measure ν and common-minorization mass δ . The finite tensor becomes a matrix-valued measure, and the lower endpoint becomes an entropy integral. This extension does not remove the statistical difficulty of estimating rich conditional report laws.

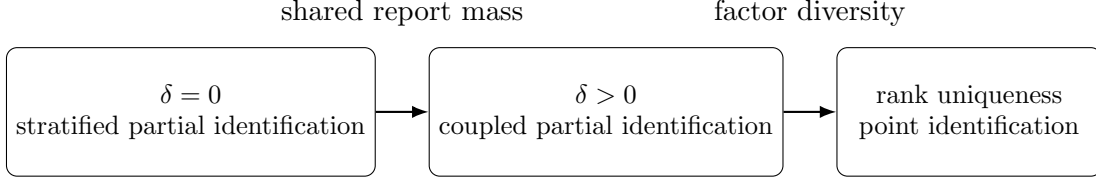


Figure 3: Three identification regimes. Common minorization prevents perfect report-indexing and induces cross-report coupling; tensor uniqueness supplies point identification under stronger rank conditions.

7 Statistical Inference

Identification treats the observed law as known. Finite samples add uncertainty in the endpoint functionals and in the compatibility restrictions. We give an interior influence-function calculation, a cluster-aware compatibility region, and a boundary-safe projection method.

7.1 Endpoint inference in finite models

Fix $C = c$ and stack the finite observed tensor into a probability vector p . For fixed channel restrictions, let D denote the linear map that decontaminates the proxy and report modes:

$$z = Dp = \text{vec}(\mathcal{Z}).$$

The sharp endpoints from [Theorem 6.2](#) are

$$\ell(p) := H_p(Y | T) - H_{Dp}(Y | J, K), \quad (59)$$

$$u(p) := H_p(Y | T) - \sum_i \gamma_i(p) H(u^i). \quad (60)$$

Assume that p and Dp lie in the relative interiors of their compatible probability simplexes.

Define

$$h_p(y, t) := -\log_2 p(y | t), \quad (61)$$

$$h_z(y, j, k) := -\log_2 z(y | j, k). \quad (62)$$

Let D^* be the adjoint of D .

Theorem 7.1 (Endpoint influence functions). *In the finite interior model, influence functions for (59) and (60) are*

$$\begin{aligned} \phi_\ell(O) &= h_p(Y, T) - H_p(Y | T) \\ &\quad - [D^* h_z(O) - H_z(Y | J, K)], \end{aligned} \quad (63)$$

$$\phi_u(O) = h_p(Y, T) - H_p(Y | T) - [c_a(Y) - \mathbb{E}c_a(Y)], \quad (64)$$

where

$$c_a(i) := \frac{H(u^i)}{1 - A}.$$

Under a symmetric outcome floor, all $H(u^i)$ are equal and

$$\phi_u(O) = h_p(Y, T) - H_p(Y | T). \quad (65)$$

The plug-in endpoints are asymptotically linear at interior laws. Repeated observations from the same subject or dialogue are aggregated at the cluster level, and the independent cluster count determines the asymptotics [Cameron et al., 2008]. Section C gives the derivative and variance calculations.

7.2 Compatibility as moment inequalities

For a sensitivity parameter $\theta = (a, b, d)$, every canonical cell is a linear moment

$$\mathcal{W}_r(\theta) = \mathbb{E}[\psi_r(O; \theta)]. \quad (66)$$

The population model is compatible exactly when all moments are nonnegative. With independent clusters $g = 1, \dots, G_n$, a cluster multiplier process approximates the joint lower tail of the studentized moments, following the simultaneous-inference logic for moment inequalities and maxima of many statistics [Andrews and Soares, 2010, Chernozhukov et al., 2013]. A parameter value is rejected when at least one canonical cell remains negative after simultaneous sampling uncertainty is included.

We report the confidence-valid feasible region

$$\hat{\Theta}_{1-\beta} := \{\theta : \theta \text{ is not rejected at level } \beta\}. \quad (67)$$

This region answers an empirical question about compatibility. Scientific calibration of a , b , and (δ, ν) requires design information or external evidence.

7.3 Confidence regions for the identified set

At boundary laws, canonical masses can vanish and the endpoint maps need not be differentiable. Let $\mathcal{C}_{1-\alpha}$ be a confidence set for the observed law with

$$\mathbb{P}\{P_O \in \mathcal{C}_{1-\alpha}\} \geq 1 - \alpha,$$

and let $\mathcal{O}_\theta$ denote the observed laws compatible with θ .

Proposition 7.2 (Confidence-set projection). *Define*

$$L_n(\theta) := \inf_{Q \in \mathcal{C}_{1-\alpha} \cap \mathcal{O}_\theta} \ell_\theta(Q), \quad (68)$$

$$U_n(\theta) := \sup_{Q \in \mathcal{C}_{1-\alpha} \cap \mathcal{O}_\theta} u_\theta(Q). \quad (69)$$

If $P_O \in \mathcal{O}_\theta$, then

$$\mathbb{P}\{\mathcal{I}_\Gamma(P_O; \theta) \subseteq [L_n(\theta), U_n(\theta)]\} \geq 1 - \alpha. \quad (70)$$

For multinomial data, $\mathcal{C}_{1-\alpha}$ can be a likelihood or ℓ_1 region. Interior influence-function intervals are shorter; projection supplies coverage at the boundary [Imbens and Manski, 2004, Chernozhukov et al., 2007, Romano and Shaikh, 2010, Kaido et al., 2019].

7.4 Rich report representations

The standard-Borel population theorem leads to a cross-fitted one-step estimator when (δ, ν) and the proxy transformation are fixed. The required nuisance functions are conditional outcome laws under the observed and canonical distributions. Under positivity and a product-rate condition, the estimator is asymptotically linear. Section C gives the adjoint operator, score, and high-level conditions. The same appendix gives a valid outer region for approximate channel structure.

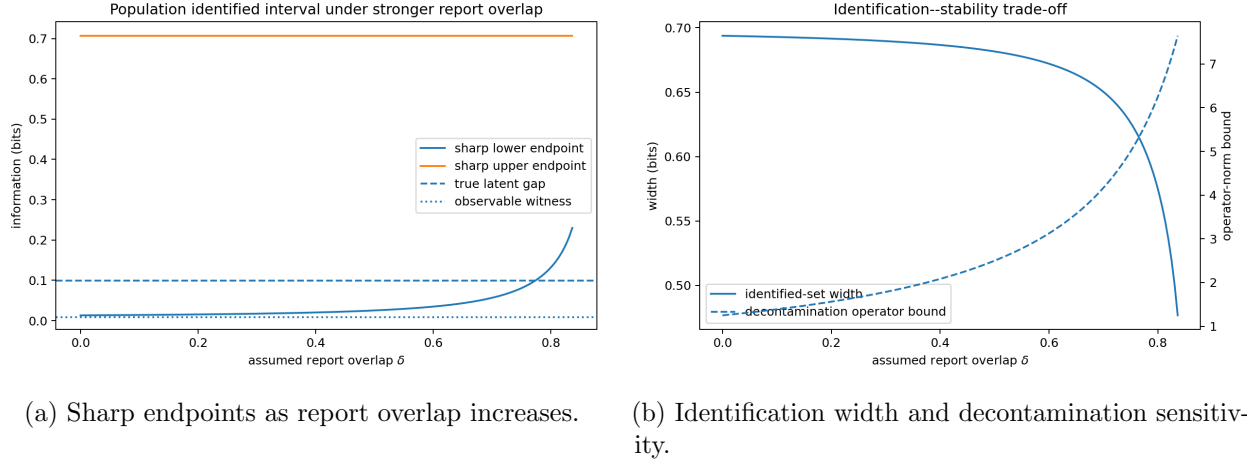


Figure 4: Population simulation. Stronger common report mass raises the lower endpoint and narrows the identified set. The inverse report channel amplifies perturbations as the restriction strengthens.

8 Simulation and Three Empirical Regimes

The simulation checks sharpness, endpoint inference, and the identification–stability trade-off. NHANES then provides a finite-table application in which the empirical estimand is the observable functional W itself. MELD supplies a model-class predictive stress test, and CrossCheck supplies a participant-clustered longitudinal robustness study.

Code, compact analysis inputs, frozen configurations, and reference outputs for all analyses in this section are available in the [public reproduction repository](#). From these inputs, the repository reruns the simulations and the downstream NHANES, MELD, and CrossCheck analyses, including the reported tables and figures. It does not reconstruct the compact inputs from excluded source-level data: NHANES minute-level accelerometry, MELD transcripts, media, and high-dimensional features, and the original CrossCheck measurements. The repository records the upstream sources, transformations, and hashes; reproducing those preprocessing stages requires obtaining the corresponding third-party data.

8.1 Simulation

We generate a binary three-view model with three latent states and

$$\alpha = 0.05, \quad \eta = 0.10, \quad \delta = 0.10.$$

The latent weights are $\pi = (0.30, 0.40, 0.30)$. Outcome, proxy, and report channels are obtained by mapping interior simplex coordinates through their floor matrices. The population witness is 0.00834 bits, the generating latent gap is 0.09990 bits, and the sharp interval at the generating value $\delta = 0.10$ is

$$[0.01404, 0.70691] \text{ bits.}$$

Under fixed (α, η) , the observed law remains compatible up to $\delta \approx 0.841$.

The observable witness, the generating gap, and the identified interval play separate roles. The first is fixed by the observed law, the second by the latent data-generating model, and the third by

Table 4: Finite-sample endpoint inference in the interior simulation. Set coverage refers to an outer confidence region for the full population identified set.

n	RMSE lower	RMSE upper	lower coverage	upper coverage	set coverage
300	0.0184	0.0104	0.957	0.947	0.996
600	0.0114	0.0065	0.947	0.943	0.989
1200	0.0076	0.0046	0.948	0.934	0.979
2400	0.0053	0.0030	0.944	0.937	0.970

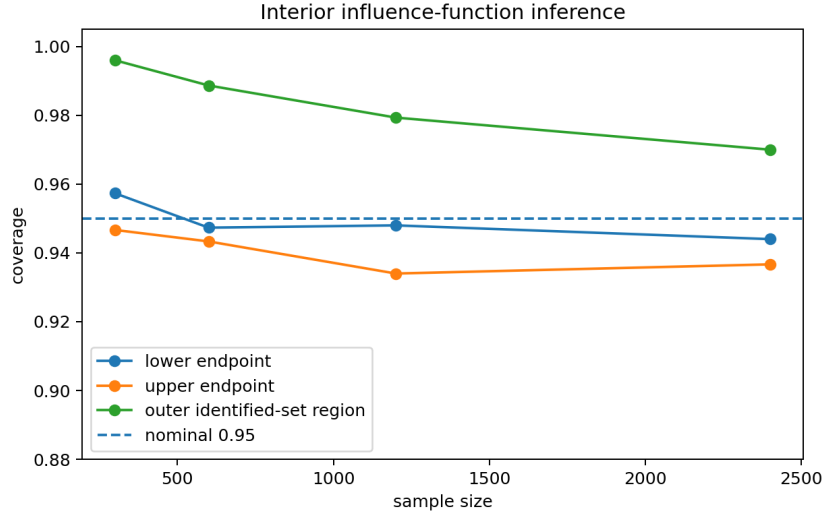


Figure 5: Coverage in the interior simulation. Pointwise endpoint intervals track the nominal level; the outer region for the full identified set is conservative.

the observed law together with the maintained channel restrictions. Figure 4b shows the trade-off between a narrower population set and a less stable decontamination map.

For inference, we draw multinomial samples at

$$n \in \{300, 600, 1200, 2400\}$$

with 1,500 repetitions per design. We compute plug-in endpoints and their interior influence-function standard errors.

The endpoint estimators approach their population values, and the outer set region is mildly conservative at these sample sizes. Boundary cases require confidence-set projection.

8.2 NHANES self-report, accelerometry, and mortality

We combine the 2003–2004 and 2005–2006 National Health and Nutrition Examination Survey (NHANES) physical-activity monitor files with questionnaire, demographic, and 2019 public-use linked-mortality records [National Center for Health Statistics, 2007, 2022, Troiano et al., 2008]. The temporal ordering is baseline report and accelerometry followed by mortality. We define T as the four-level PAQ180 self-report, M as quantile bins of valid-wear accelerometer counts per minute (CPM), Y as death within ten years of the examination, and C as sex crossed with age below or at least 60 years.

Table 5: Survey-weighted observable witness in NHANES. The primary row is ten-year mortality with four CPM bins. Other rows are prespecified representation and horizon checks. Permutation tests are not multiplicity-adjusted.

Horizon	CPM bins	deaths	plug-in $\widehat{W}$	bias-corrected	95% CI
10 years	2	903	0.01130	0.01041	[0.00650, 0.01374]
10 years	3	903	0.02011	0.01817	[0.01400, 0.02218]
10 years	4	903	0.02984	0.02753	[0.02194, 0.03272]
5 years	2	405	0.00916	0.00872	[0.00660, 0.01084]
5 years	3	405	0.01352	0.01256	[0.00933, 0.01553]
5 years	4	405	0.02006	0.01870	[0.01472, 0.02240]

Minute counts are processed with the National Cancer Institute conventions used for these NHANES cycles: at least four valid days, at least 600 wear minutes per valid day, and 60 consecutive zero-count minutes as nonwear with the declared short-interruption tolerance. The analysis includes 6,565 adults with complete report, accelerometer, mortality, and weight information, including 903 deaths within ten years. We pool the cycles with WTMEC2YR/2 and resample primary sampling units within cycle-specific survey strata. Because monitor validity and complete-case inclusion can change the represented population, we interpret W as a survey-weighted functional of the analyzed cohort rather than automatically generalizing it to all US adults.

Unlike the MELD score comparison below, the finite table directly targets

$$\widehat{W} = I_{\widehat{P}}(Y; M \mid T, C). \quad (71)$$

The primary specification uses four CPM bins. Two- and three-bin proxies form a representation ladder, and five-year mortality is a prespecified horizon check. Table 5 reports plug-in and bootstrap bias-corrected estimates. Confidence intervals use 1,000 survey-PSU bootstrap draws; conditional permutation is performed within report, context, and cycle cells.

The primary witness is 0.02984 bits (95% CI [0.02194, 0.03272]; conditional-permutation $p = 0.000999$). Thus objective accelerometer activity contains statistically robust mortality-relevant information beyond the declared self-report and context representation. We do not interpret this quantity as a causal effect or as the amount of information in a uniquely defined latent health state. Under the unrestricted statistical extension class, it instantiates the sharp benchmark [0.02984, 0.34118] bits; the upper endpoint is the observed conditional-entropy ceiling, not an additional empirical discovery.

The positive-floor analysis exposes a boundary problem rather than resolving it by smoothing. In the frozen $4M \times 4T \times 4C$ table, empirical empty cells make every strictly positive grid point plug-in incompatible. A boundary-safe, PSU-block ℓ_1 outer screen cannot reject any of the 216 grid points, but it certifies none because its context-specific radii are wide. The data therefore establish a positive observable witness but do not calibrate a nonzero channel floor in the canonical representation. Post-canonical coarsening diagnostics, reported in Section D.3, show that positive-floor plug-in intervals can appear when the alphabet or conditioning set is reduced; these are representation-sensitivity results, not replacements for the primary analysis.

8.3 MELD next-turn sentiment

MELD contains multimodal, multi-party conversations with utterance-level emotion and sentiment labels [Poria et al., 2019]. We form 2,330 adjacent-turn pairs from 268 dialogues and predict the

Table 6: Dialogue-grouped incremental prediction of MELD next-turn sentiment. Confidence intervals use dialogue-cluster multiplier inference.

Modality	gain (bits)	95% lower	95% upper
Audio	0.00293	-0.00048	0.00633
Visual	-0.00125	-0.00626	0.00375
Audio+visual	0.00103	-0.00427	0.00633

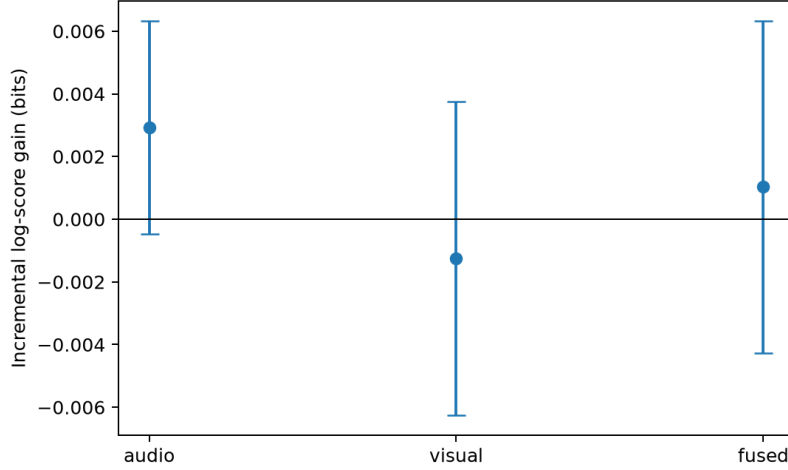


Figure 6: Cross-fitted non-report gains for MELD next-turn sentiment under the frozen canonical analysis. All full-sample dialogue-cluster intervals include zero.

three-class sentiment label of the next utterance.

For the current utterance, the report representation R contains 32 transcript SVD components. The non-report representations contain 16 audio principal components, 32 visual principal components, or their concatenation. All transformations are fitted on the MELD training split. Context records whether the current speaker is one of the six major characters and whether the next utterance has the same speaker. The scientific target remains representation-relative, but the computed quantity is a model-class score gain rather than W itself.

We estimate the fitted modality gain with multinomial ridge logistic regression. Five outer folds use stratified grouping by dialogue. Four grouped folds within each outer training sample select the ridge penalty from a fixed grid. For a declared ridge-logistic model class $\mathcal{F}$, the out-of-fold gain is

$$\hat{\Delta}_{\mathcal{F},M} = \frac{1}{n} \sum_{i=1}^n \log_2 \frac{\hat{p}_{-i}(Y_i | R_i, M_i, C_i)}{\hat{p}_{-i}(Y_i | R_i, C_i)}. \quad (72)$$

Dialogue-cluster Rademacher multipliers provide confidence intervals for the mean score difference.

The unconditional outcome entropy is 1.4982 bits. The text-only out-of-fold log loss is 1.4959 bits. Table 6 and Figure 6 report the incremental gains.

The full-sample analysis finds no stable out-of-dialogue model-class modality gain. Negative fitted gains, which mutual information cannot take, make clear that $\hat{\Delta}_{\mathcal{F},M}$ is not being presented as a direct estimator of W . Repeated splits, a same-speaker subgroup, a representation ladder, and exploratory model-based endpoint calculations are reported in Section D. The same-speaker fused estimate is positive in an unadjusted exploratory interval; it is not part of the primary claim.

Reproducibility. The canonical analysis freezes input hashes, software versions, outer and inner fold manifests, random seeds, the ridge grid, and multiplier settings. The release contains out-of-fold predictions and output hashes. Two complete runs agree on all 111 compared numerical cells.

The MELD result is task- and model-class-specific: current audio and visual representations do not provide a robust full-sample predictive gain for next-turn sentiment after continuous report information and dialogue structure are included. It does not establish that the modalities contain no additional observable task information.

8.4 CrossCheck longitudinal robustness

CrossCheck contains repeated ecological momentary assessments (EMAs) and aligned passive mobile-sensing summaries [Adler et al., 2022]. We form 5,607 adjacent-assessment pairs from 60 participants. The report T is the current negative-symptom EMA tertile, M is a tertile of a four-sensor composite, Y is the next EMA tertile within one to seven days, and C records above- or below-median sensing quality. Participants receive equal total weight, and uncertainty is resampled at the participant level.

The plug-in witness is 0.01037 bits and the bootstrap bias-corrected estimate is 0.00121 bits. The participant-bootstrap 95% interval $[-0.01182, 0.01037]$ includes zero, although the conditional-permutation test gives $p = 0.02797$. We therefore treat CrossCheck as inconclusive: the point estimate and permutation diagnostic indicate dependence, but the independent-participant information is insufficient to establish a robust non-report witness. This contrast with NHANES illustrates why repeated observations cannot be treated as independent evidence.

9 Discussion

The paper studies the value of an unobserved experiment from a fixed observed law. The observable proxy value $I(Y; M | T, C)$ is identified. The latent value $I(Y; E | T, C)$ ranges over extensions that reproduce the same observables and satisfy the maintained measurement restrictions. Blackwell ordering explains the baseline inequality; refinement and merging determine the sharp endpoints under restricted channel geometry.

9.1 Statistical extensions and scientific states

The identified set is defined over statistical latent extensions. A scientific interpretation of E may require additional structure, including temporal priority, nonanticipation, exclusion of direct proxy effects, or external validation of the measurement channels. These conditions define a smaller extension class and can tighten the bounds. The unrestricted endpoint constructions establish what the observed law and local independence alone cannot exclude.

This distinction matters most at the unrestricted upper endpoint, where a saturated extension can include the realized outcome. The construction is a sharpness device for the statistical model class. Applications that require a pre-outcome state should impose that requirement explicitly rather than attach it to the notation after identification.

The NHANES ordering makes the scientific interpretation more plausible without identifying a unique state: self-report and accelerometry are measured at baseline and mortality occurs later. The positive witness therefore concerns information in a specific objective measurement before the outcome. It still does not identify the content of E , establish a causal effect of activity, or justify an unrestricted endpoint construction as a literal physiological mechanism.

9.2 Calibration and empirical design

Outcome floors, proxy floors, report minorization, and latent cardinality are sources of identifying content. Their mathematical effects are transparent: outcome floors lower the upper endpoint, proxy floors raise the lower endpoint, common report mass couples strata, and cardinality limits refinement. Their numerical values require calibration from design knowledge, validation samples, repeated measurements, or sensitivity analysis.

Observable compatibility and substantive calibration answer separate questions. A compatibility test determines whether a proposed parameter value can generate the data. External evidence determines whether the value is scientifically credible. Strong applications should combine both. Temporal ordering, independently collected outcomes, repeated proxy measurements, and negative controls also support the conditional measurement structure.

Probability floors are representation-dependent. Splitting a proxy category changes the numerical meaning of b_m and can create sparse empirical cells; coarsening a report or context changes the estimand and the compatibility geometry. The NHANES proxy ladder therefore assesses observable information retained by alternative CPM discretizations, while the post-canonical positive-floor calculations are reported only as representation-sensitivity diagnostics. Neither a nonrejected floor nor a compatible coarsening supplies external calibration.

9.3 Implications for language-based systems

The framework concerns report generation. Text as a storage format can encode arbitrary digital measurements; the relevant channel maps a state into a report available to the system. Sufficiency is defined relative to the outcome. A report can support one decision and leave information for another.

The analysis also separates observation from computation. Model capacity can reduce representation and prediction errors in (11). It does not recover information absent from a fixed observation. A system that asks questions or requests new measurements changes the acquisition channel and belongs to an active design.

Empirical studies usually condition on a representation $R = f(T)$. [Theorem 3.4](#) shows that information discarded by R is added to the representation-relative gap. Richer language models can reduce this component, while sufficiency of the original report remains a property of the task and the report-generating process.

9.4 Scope and next steps

The core population results use unrestricted latent support. A binding cardinality restriction recouples the channels through nonnegative factorization, and classical rank conditions can lead to point identification. The standard-Borel extension shows that finite report alphabets are not required at the population level; high-dimensional estimation still depends on conditional-distribution learning, positivity, and decontamination stability.

The empirical studies separate three regimes. NHANES directly estimates a positive finite-table witness with a future outcome, MELD reports a model-class predictive gain that cannot be equated with W , and CrossCheck shows how participant-level uncertainty can leave a positive point estimate inconclusive. In the canonical NHANES representation, conservative boundary inference is too weak to calibrate a nonzero channel floor. External validation of outcome, proxy, and report-channel restrictions remains necessary for a scientifically interpretable restricted identified set. Active elicitation introduces adaptive questions and state intervention and calls for a controlled partially observed process.

9.5 Conclusion

Multimodal gain and latent report insufficiency are distinct statistical objects. The first measures the value of a particular observed proxy. The second asks for the value of an unobserved state compatible with the observed law. Restricted channel geometry turns that question into a sharp partial-identification problem with observable compatibility conditions and finite-sample inference.

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A Boundary Regimes and General Report Spaces

This appendix records three extensions that mark the boundary of the main partial-identification regime: bounded latent cardinality, tensor-unique point identification, and standard-Borel report spaces.

A.1 Finite latent cardinality

In the finite cross-report simplex model, write

$$q_e = Ua_e, \quad r_e = Vb_e, \quad s_e = Sc_e.$$

Then

$$\mathcal{W} = \sum_{e=1}^K \pi_e a_e \otimes b_e \otimes c_e. \quad (73)$$

Theorem A.1 (Latent cardinality and nonnegative CP rank). *For a compatible finite-view model,*

$$\boxed{\min\{|\mathcal{E}| : \text{the model generates } \mathcal{P}\} = \text{rank}_+^{\text{CP}}(\mathcal{W})}. \quad (74)$$

Equivalently, $|\mathcal{E}| \leq K$ is feasible if and only if $\text{rank}_+^{\text{CP}}(\mathcal{W}) \leq K$.

In the two-view simplex model, the corresponding condition is $\text{rank}_+(\Omega_x) \leq K$. Exact nonnegative matrix factorization is NP-hard when the target rank is part of the input [Vavasis, 2009]; identifiability of its factors requires additional geometric or structural conditions [Fu et al., 2018, Gillis and Rajkó, 2023]. Fixed-rank and small-scale instances can be handled with specialized algebraic or global-optimization methods.

For outcome vertex i , let

$$W_i = [\mathcal{W}_{ijk}]_{j,k}$$

be the corresponding proxy-report slice.

Theorem A.2 (Minimum cardinality for upper-endpoint attainment). *Assume the outcome vertices u^i are distinct and Shannon entropy is used. With $\text{rank}_+(0) = 0$, the minimum number of positive-weight latent states required to attain (54) is*

$$\boxed{K_{\min}^{\text{upper}} = \sum_{i:\gamma_i>0} \text{rank}_+(W_i)}. \quad (75)$$

Classical tensor-uniqueness results give the point-identification boundary. Let Q , R , and S_E denote the factor matrices with columns q_e , r_e , and s_e .

Corollary A.3 (Point identification under tensor uniqueness). *If K is known and the Kruskal ranks satisfy*

$$k_Q + k_R + k_{S_E} \geq 2K + 2, \quad (76)$$

then, under the usual nondegeneracy conditions, the latent-class decomposition is unique up to label permutation. Consequently, Γ_c is point identified.

This corollary applies tensor-identification theory [Kruskal, 1977, Allman et al., 2009, Anandkumar et al., 2014]. Common minorization narrows a partially identified extension class; tensor uniqueness identifies a particular decomposition.

A.2 Standard-Borel report spaces

Let $(\mathcal{T}, \mathcal{F})$ be a standard Borel report space and fix a reference report distribution ν . Assume

$$s_e(A) \geq \delta \nu(A) \quad \text{for every } e \text{ and } A \in \mathcal{F}, \quad (77)$$

with $0 \leq \delta < 1$. Equivalently,

$$s_e = \delta \nu + (1 - \delta) \tilde{s}_e.$$

The largest common mass, optimized over ν , is the Doeblin coefficient of the report kernel [Makur and Singh, 2024]. The theorem treats (δ, ν) as fixed calibration or sensitivity input.

For measurable $A \subseteq \mathcal{T}$, define the $L \times J$ matrix-valued measure

$$P(A) = [P(Y = y, M = m, T \in A \mid c)]_{y,m}$$

and

$$C(A) := U^{-1} P(A) V^{-\top}, \quad (78)$$

$$\mathcal{W}_{ij}(A) := \frac{C_{ij}(A) - \delta \nu(A) C_{ij}(\mathcal{T})}{1 - \delta}. \quad (79)$$

Theorem A.4 (Measure-valued compatibility and sharp endpoints). *With unrestricted latent state space, the model is compatible if and only if every $\mathcal{W}_{ij}$ in (79) is a nonnegative finite measure.*

For the lower endpoint, define

$$Z(A) := P(A) V^{-\top}, \quad (80)$$

$$\tilde{Z}(A) := \frac{Z(A) - \delta \nu(A) Z(\mathcal{T})}{1 - \delta}. \quad (81)$$

For each proxy coordinate j , let

$$\lambda_j(A) := \sum_y \tilde{Z}_{yj}(A)$$

and define Radon–Nikodym derivatives

$$q_y^j(t) := \frac{d\tilde{Z}_{yj}(t)}{d\lambda_j}.$$

Then

$$\boxed{\underline{\Gamma}_c = H(Y \mid T, c) - \sum_j \int_{\mathcal{T}} H(q^j(t)) \lambda_j(dt).} \quad (82)$$

The sharp upper endpoint is

$$\boxed{\overline{\Gamma}_c = H(Y \mid T, c) - \sum_i \gamma_i H(u^i).} \quad (83)$$

The result is an explicit population characterization. Estimation on a rich report space requires conditional-distribution learning and control of positivity and decontamination stability. The measure transformation is related to mutual-contamination demixing [Katz-Samuels et al., 2019]; the estimand here is the value of a latent experiment.

B Proofs of the Identification Results

B.1 Unrestricted benchmark and report representations

Proof of Theorem 3.3. Condition on C throughout. Under $Y \perp\!\!\!\perp M \mid E, T, C$,

$$\begin{aligned} I(Y; E, M \mid T, C) &= I(Y; E \mid T, C) + I(Y; M \mid E, T, C) \\ &= I(Y; E \mid T, C), \end{aligned}$$

while the reverse chain rule gives

$$I(Y; E, M \mid T, C) = I(Y; M \mid T, C) + I(Y; E \mid T, M, C).$$

Hence

$$\Gamma = I(Y; M \mid T, C) + I(Y; E \mid T, M, C) \geq I(Y; M \mid T, C).$$

The lower endpoint is attained by setting $E = (T, M)$ and

$$P(Y = y \mid E = (t, m), C = c) = P_O(y \mid t, m, c).$$

Then T and M are deterministic given (E, C) and the observed law is preserved.

The upper bound follows from

$$I(Y; E \mid T, C) \leq H(Y \mid T, C).$$

It is attained by $E = (T, M, Y)$, with all three observed variables deterministic given (E, C) . To obtain any convex combination of the endpoints, let B be independent of (Y, M, T, C) and, conditional on $B = b$, use an endpoint-attaining latent extension. The augmented state (B, E_B) preserves the observed law and its conditional information is the corresponding convex combination. Thus every value in the stated interval is attainable. $\square$

Proof of Theorem 3.4. Because $R = f(T)$,

$$\begin{aligned} I(Y; E, T \mid R, C) &= I(Y; T \mid R, C) + I(Y; E \mid T, R, C) \\ &= I(Y; T \mid R, C) + I(Y; E \mid T, C). \end{aligned}$$

The same quantity can be expanded in the other order:

$$\begin{aligned} I(Y; E, T \mid R, C) &= I(Y; E \mid R, C) + I(Y; T \mid E, R, C) \\ &= I(Y; E \mid R, C), \end{aligned}$$

where the final equality uses $Y \perp\!\!\!\perp T \mid E, C$. Equating the two expressions proves the result. $\square$

B.2 Refinement–merging principle

Proof of Theorem 4.1. Take any admissible latent decomposition. Since $r_e \in \mathcal{R}$, choose coefficients $\beta_{ej} \geq 0$ such that

$$r_e = \sum_{j=1}^B \beta_{ej} r^j, \quad \sum_j \beta_{ej} = 1.$$

Define

$$z_j := \sum_e \pi_e \beta_{ej} q_e, \quad \lambda_j := \mathbf{1}^\top z_j = \sum_e \pi_e \beta_{ej}.$$

Convexity of $\mathcal{Q}$ gives $z_j \in \text{cone}(\mathcal{Q})$, and

$$P_x = \sum_e \pi_e q_e r_e^\top = \sum_j z_j (r^j)^\top.$$

For each positive λ_j , concavity of Φ yields

$$\tilde{\Phi}(z_j) = \lambda_j \Phi\left(\frac{\sum_e \pi_e \beta_{ej} q_e}{\lambda_j}\right) \geq \sum_e \pi_e \beta_{ej} \Phi(q_e).$$

Summing over j gives

$$\sum_j \tilde{\Phi}(z_j) \geq \sum_e \pi_e \Phi(q_e).$$

Thus every latent decomposition maps to a feasible point of the stated program with weakly larger conditional Bayes risk. Conversely, any feasible $z_1, \dots, z_B$ defines an admissible model with one state per positive z_j : use weight λ_j , outcome channel z_j/λ_j , and proxy channel r^j . This model attains the program objective. At most B states are needed. $\square$

Proof of Theorem 4.2. Write every $q_e \in \mathcal{Q}$ as

$$q_e = \sum_{i=1}^A \alpha_{ei} u^i, \quad \alpha_{ei} \geq 0, \quad \sum_i \alpha_{ei} = 1.$$

Define

$$m_i := \sum_e \pi_e \alpha_{ei} r_e.$$

Then $m_i \in \text{cone}(\mathcal{R})$ and

$$P_x = \sum_i u^i m_i^\top.$$

Concavity gives

$$\Phi(q_e) \geq \sum_i \alpha_{ei} \Phi(u^i),$$

so

$$\sum_e \pi_e \Phi(q_e) \geq \sum_i (\mathbf{1}^\top m_i) \Phi(u^i).$$

Any feasible collection m_i yields a latent model with weight $\gamma_i = \mathbf{1}^\top m_i$, outcome channel u^i , and proxy channel m_i/γ_i . Hence the lower bound on conditional Bayes risk is attainable with at most A states. If $\mathcal{R}$ is described by linear inequalities, membership of m_i in its conic hull is linear after homogenization, and the objective is linear. $\square$

B.3 Simplex channels

Proof of Theorem 5.1. Every admissible latent model has unique extreme coordinates $a_e \in \Delta_L$ and $b_e \in \Delta_J$. Therefore

$$P_x = U \left(\sum_e \pi_e a_e b_e^\top \right) V^\top,$$

which proves that the identified matrix $\Omega_x = U^{-1} P_x V^{-\top}$ is entrywise nonnegative. Conversely, if $\Omega_x \geq 0$, its entries sum to one because the columns of U and V are probability vectors. A latent state (i, j) with weight $(\Omega_x)_{ij}$, outcome channel u^i , and proxy channel v^j generates P_x .

The lower formula follows from [Theorem 4.1](#). The j th conic outcome vector is

$$z_j = P_x V^{-\top} e_j,$$

and its mass and normalized outcome channel are λ_j and $\bar{q}_j$. The upper formula follows from [Theorem 4.2](#), because the i th conic proxy vector is the transpose of the i th row of $U^{-1}P_x$. The cell construction and the endpoint constructions can be mixed with an independent indicator, so unrestricted latent cardinality gives the full interval between the endpoints. $\square$

Proof of Theorem 5.2. The vertices of $\mathcal{Q}_a$ are $u^i = a + (1 - A)e_i$, and

$$U = (1 - A)I + a\mathbf{1}^\top, \quad U^{-1} = \frac{I - a\mathbf{1}^\top}{1 - A}.$$

The inverse identity follows by direct multiplication and $\mathbf{1}^\top a = A$. The analogous formulas hold for V .

Using $p_Y = P_x \mathbf{1}$ and $p_M = P_x^\top \mathbf{1}$,

$$U^{-1}P_x V^{-\top} = \frac{P_x - ap_M^\top - p_Y b^\top + ab^\top}{(1 - A)(1 - B)},$$

which gives (33). Further,

$$P_x V^{-\top} e_m = \frac{P_x e_m - b_m p_Y}{1 - B}.$$

Its mass is $(p_m - b_m)/(1 - B)$ and its normalized vector is (35). The row sums of $U^{-1}P_x$ give (36). Substitution into [Theorem 5.1](#) proves the endpoint formulas. Under unrestricted latent cardinality the identified set is the full interval between them. $\square$

Proof of Theorem 5.3. The lower-endpoint construction gives a Markov chain $Y - J^\star - M$ with

$$\underline{\Gamma}_x = I(Y; J^\star \mid X = x).$$

Data processing implies $I(Y; M \mid x) \leq I(Y; J^\star \mid x)$. Suppose equality holds. For the Markov chain, equality is equivalent to

$$I(Y; J^\star \mid M, x) = 0.$$

Together with $Y \perp\!\!\!\perp M \mid J^\star, x$, this gives

$$P(Y \mid J^\star = j, x) = P(Y \mid M = m, x)$$

for every active j and every m with positive joint probability. Each proxy vertex has entries

$$v_m^j = b_m + (1 - B)\mathbf{1}\{m = j\}.$$

Since $b_m > 0$ for all m , every active j connects to every m . Hence the conditional law of Y is the same for all active values of $J^\star$, which implies $I(Y; J^\star \mid x) = 0$. This contradicts $I(Y; M \mid x) > 0$. The inequality is therefore strict. $\square$

B.4 Cross-report identification

Proof of Theorem 6.1. Let the latent state space be the disjoint union of the stratum-specific state spaces:

$$\mathcal{E}' = \{(t, h) : t \in \mathcal{T}, h \in \mathcal{H}_t\}.$$

Given $C = c$, assign

$$P(E' = (t, h) \mid c) = P(T = t \mid c) \pi_{h|t,c},$$

let T be deterministic given $E' = (t, h)$, and set

$$P(Y \mid E' = (t, h), c) = q_{h,t,c}, \quad P(M \mid E' = (t, h), c) = r_{h,t,c}.$$

The three observed variables are conditionally independent given (E', C) , and marginalizing E' recovers every conditional table in (42). Thus unrestricted state refinement absorbs the report label. $\square$

Proof of Theorem 6.2. Write

$$q_e = U a_e, \quad r_e = V b_e, \quad s_e = S c_e,$$

with a_e, b_e, c_e probability vectors. Then

$$\mathcal{W} = \mathcal{P} \times_1 U^{-1} \times_2 V^{-1} \times_3 S^{-1} = \sum_e \pi_e a_e \otimes b_e \otimes c_e,$$

so compatibility implies $\mathcal{W} \geq 0$. Conversely, if $\mathcal{W} \geq 0$, use a latent state (i, j, k) with weight $\mathcal{W}_{ijk}$ and channel triple (u^i, v^j, w^k) , where w^k is the k th column of S . This cell construction recovers $\mathcal{P}$.

For the lower endpoint, first refine every proxy coordinate b_e and report coordinate c_e to simplex vertices. The resulting model has states indexed by the original state and a pair (j, k) and preserves $\mathcal{P}$ and the average outcome entropy. Merge all states sharing (j, k) . Their total outcome vector is

$$z^{jk} = \sum_e \pi_e b_{ej} c_{ek} q_e,$$

which is exactly the (j, k) slice of $\mathcal{Z}$. Its mass is λ_{jk} and its normalized channel is q^{jk} . Concavity of entropy shows that merging weakly increases average conditional entropy, hence weakly lowers Γ_c . Conversely, the merged model is admissible and attains the displayed value. It uses one state for each positive λ_{jk} .

For the upper endpoint, refine every q_e to the outcome vertices u^i . Entropy concavity gives the universal lower bound

$$\sum_e \pi_e H(q_e) \geq \sum_i \gamma_i H(u^i).$$

Full compatibility provides a construction attaining this bound: for each i , decompose the nonnegative slice $\mathcal{W}_{i..}$ into cells, or into any nonnegative rank factorization, and attach outcome channel u^i . The total mass on outcome vertex i is the i th marginal of $\mathcal{W}$, equal to $(P(Y = i \mid c) - a_i)/(1 - A)$. Thus the upper formula is sharp. Independent mixing of endpoint-attaining extensions fills the interval when latent cardinality is unrestricted. $\square$

Proof of Theorem 6.3. When $d = 0$, $S = I$ and the report extreme coordinate is the observed report value. Hence

$$z^{jt} = P(T = t \mid c) z_{j|t,c},$$

where $z_{j|t,c}$ is the lower-endpoint conic vector for the conditional (Y, M) table in stratum (t, c) . Entropy perspective is homogeneous, so summing the local lower objectives over t gives (56). $\square$

Proof of Theorem A.1. Every K -state latent model yields the nonnegative CP decomposition

$$\mathcal{W} = \sum_{e=1}^K \pi_e a_e \otimes b_e \otimes c_e,$$

so $\text{rank}_+^{\text{CP}}(\mathcal{W}) \leq K$. Conversely, take any nonnegative CP decomposition of $\mathcal{W}$ with R terms. Normalize each nonzero factor to sum to one and absorb the three factor masses into a scalar weight. Since $\mathcal{W}$ has total mass one, the normalized weights sum to one. Mapping the normalized factors through U, V, S gives an R -state admissible latent model for $\mathcal{P}$. Minimizing over representations proves the equality. $\square$

Proof of Theorem A.2. For any state, write $q_e = \sum_i a_{ei} u^i$. Strict concavity of Shannon entropy gives

$$H(q_e) \geq \sum_i a_{ei} H(u^i),$$

with equality only when all positive a_{ei} correspond to the same vertex. Since the global upper endpoint requires equality after averaging, every positive-weight state must have $q_e = u^i$ for one i .

Fix i . The states with outcome channel u^i induce a nonnegative factorization

$$W_i = \sum_{e: q_e = u^i} \pi_e b_e c_e^T.$$

At least $\text{rank}_+(W_i)$ terms are necessary. Conversely, a minimum nonnegative factorization of each nonzero W_i supplies exactly $\text{rank}_+(W_i)$ latent states with outcome channel u^i . Combining the factorizations over i generates $\mathcal{W}$ and attains the upper endpoint. Summing the minimum term counts proves the result. $\square$

B.5 Standard-Borel report spaces

Proof of Theorem A.4. For an admissible latent model, write $q_e = U a_e$, $r_e = V b_e$, and

$$s_e = \delta \nu + (1 - \delta) \tilde{s}_e.$$

Then

$$C_{ij}(A) = \sum_e \pi_e a_{ei} b_{ej} s_e(A),$$

and therefore

$$\mathcal{W}_{ij}(A) = \sum_e \pi_e a_{ei} b_{ej} \tilde{s}_e(A),$$

which is a nonnegative finite measure.

Conversely, suppose every $\mathcal{W}_{ij}$ is nonnegative. Their total masses sum to one. Use the standard-Borel latent space

$$\mathcal{E} = \{1, \dots, L\} \times \{1, \dots, J\} \times \mathcal{T}$$

with mixing measure $\mathcal{W}_{ij}(dt)$, outcome channel u^i , proxy channel v^j , and report kernel

$$s_{ijt} = \delta \nu + (1 - \delta) \delta_t.$$

For every measurable A the generated observable matrix is

$$\begin{aligned} \sum_{ij} \int u^i(v^j)^\top [\delta\nu(A) + (1 - \delta)\mathbf{1}\{t \in A\}] \mathcal{W}_{ij}(\mathrm{d}t) &= U[\delta\nu(A)\mathcal{W}(\mathcal{T}) + (1 - \delta)\mathcal{W}(A)]V^\top \\ &= UC(A)V^\top = P(A). \end{aligned}$$

This proves compatibility.

For the lower endpoint, proxy refinement and report-kernel refinement express the residual vector measures as

$$\tilde{Z}_{\cdot j}(A) = \sum_e \pi_e b_{ej} q_e \tilde{s}_e(A).$$

The scalar measure $\lambda_j = \sum_y \tilde{Z}_{yj}$ dominates every component. Standard-Borel disintegration gives measurable Radon–Nikodym derivatives $q^j(t)$. They are local convex combinations of admissible q_e , hence belong to $\mathcal{Q}_a$ almost everywhere. Merging states at each (j, t) raises average entropy and yields

$$\sum_j \int H(q^j(t)) \lambda_j(\mathrm{d}t).$$

The construction using latent state (j, t) , outcome channel $q^j(t)$, proxy vertex v^j , and report kernel $\delta\nu + (1 - \delta)\delta_t$ attains the value.

For the upper endpoint, refine outcome channels to u^i . The same concavity argument as in the finite model gives the lower bound $\sum_i \gamma_i H(u^i)$ on average conditional entropy. The measure-valued cell construction based on $\mathcal{W}_{ij}(\mathrm{d}t)$ attains it. This proves both sharp endpoint formulas. $\square$

C Details for Statistical Inference

C.1 Influence-function calculations

For a finite joint law $p(y, x)$ in the relative interior, the conditional entropy functional

$$\mathbf{H}(p) = H_p(Y \mid X)$$

has pathwise derivative

$$\dot{\mathbf{H}}_p[h] = \sum_{y,x} h(y, x) \{-\log_2 p(y \mid x)\}.$$

Since every score direction has total mass zero, an influence function is

$$-\log_2 p(Y \mid X) - H_p(Y \mid X).$$

For the lower endpoint, $z = Dp$ and

$$\ell(p) = \mathbf{H}_{YT}(p) - \mathbf{H}_{YJK}(Dp).$$

The chain rule for linear maps gives

$$\dot{\ell}_p[h] = \langle h, h_p \rangle - \langle Dh, h_z \rangle = \langle h, h_p - D^* h_z \rangle.$$

Centering the gradient under p gives (63). The upper calculation follows from the linearity of $\gamma_i(p)$ in the Y marginal.

For independent clusters $g = 1, \dots, G_n$, define

$$\Phi_g = \sum_{i \in g} \phi(O_i).$$

Under standard cluster asymptotics, the variance of the sample mean is estimated by

$$\widehat{V}_{\text{cl}} = \frac{G_n}{G_n - 1} \frac{1}{n^2} \sum_{g=1}^{G_n} \left(\widehat{\Phi}_g - \overline{\Phi} \right)^2.$$

All sample splitting and resampling in clustered data occur at the cluster level.

C.2 Simultaneous compatibility inference

Let R be the number of canonical cells and

$$\widehat{\mu}_r = \frac{1}{n} \sum_i \psi_r(O_i; \theta).$$

Draw independent multipliers ξ_g with mean zero and variance one and compute

$$Z_r^* = \frac{1}{\widehat{\sigma}_r \sqrt{G_n}} \sum_{g=1}^{G_n} \xi_g \left\{ \widehat{S}_{gr} - \overline{S}_r \right\}.$$

A critical value for $\max_r (-Z_r^*)$ yields simultaneous one-sided lower bands. The model is rejected when at least one canonical moment has an upper confidence bound below zero.

C.3 Confidence-set projection at the boundary

Suppose P_O has finite support of size d and $\widehat{p}$ is its empirical distribution. An ℓ_1 confidence set can be defined as

$$\mathcal{C}_{1-\alpha} = \{q \in \Delta_d : \|q - \widehat{p}\|_1 \leq \epsilon_{n,\alpha}\},$$

where $\epsilon_{n,\alpha}$ is chosen from a multinomial concentration inequality. Projection through (68)–(69) gives finite-sample set coverage when canonical masses vanish. A likelihood-ratio confidence set can be used in the same way.

For a fast conservative calculation, let D be the decontamination operator and define

$$\kappa_D = \|D\|_{\text{TV} \rightarrow \text{TV}}.$$

If $\text{TV}(P, Q) \leq \epsilon$, then $\text{TV}(DP, DQ) \leq \kappa_D \epsilon$. Combining this inequality with a conditional-entropy continuity modulus f_L gives

$$|\ell(P) - \ell(Q)| \leq f_L(\epsilon) + f_L(\min\{1, \kappa_D \epsilon\}).$$

This supplies a closed-form outer enlargement of a plug-in endpoint.

C.4 Cross-fitted general-report estimator

Consider the standard-Borel model in [Section A.2](#), with fixed (δ, ν) and fixed proxy matrix V . The lower decontamination map $\mathcal{D}$ is linear. For a test function $h(y, j, t)$, its adjoint is

$$\mathcal{D}^*h(y, m, t) = \frac{1}{1-\delta} \sum_j (V^{-1})_{jm} \left[h(y, j, t) - \delta \int h(y, j, \tau) \nu(d\tau) \right]. \quad (84)$$

Let $Q = \mathcal{D}(P)$ and define

$$h_P(y, t) = -\log_2 P(y \mid t, c), \quad h_Q(y, j, t) = -\log_2 Q(y \mid j, t, c).$$

The lower endpoint has influence function

$$\phi_\ell(O) = [h_P(Y, T) - H_P(Y \mid T, c)] - [\mathcal{D}^*h_Q(O) - H_Q(Y \mid J, T, c)]. \quad (85)$$

For fold k , fit the nuisance conditional laws outside the fold and evaluate

$$\widehat{\ell}_{\text{CF}} = \frac{1}{n} \sum_{i=1}^n \left[-\log_2 \widehat{P}_{-k(i)}(Y_i \mid T_i, C_i) - \mathcal{D}^*\{-\log_2 \widehat{Q}_{-k(i)}\}(O_i) \right]. \quad (86)$$

Under positivity, square integrability, and a product-rate condition that makes the score remainder $o_p(n^{-1/2})$,

$$\sqrt{n}(\widehat{\ell}_{\text{CF}} - \ell) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_\ell(O_i) + o_p(1).$$

Cross-fitting and orthogonal score arguments follow the general semiparametric strategy in [Chernozhukov et al. \[2018\]](#). Clustered observations require cluster-level folds and variance estimation. If ν is estimated from the analysis sample, its influence contribution must be included.

C.5 A valid outer region under approximate structure

Suppose each approximate latent channel admits a reference exact channel and the resulting observable law Q satisfies $\text{TV}(P_O, Q) \leq \rho$. Let $\Omega_L(\rho)$ bound the difference between the associated information functionals. Define

$$\begin{aligned} L_\rho(P_O) &= \inf_{Q \in \mathcal{O}_0: \text{TV}(Q, P_O) \leq \rho} \Gamma_0(Q) - \Omega_L(\rho), \\ U_\rho(P_O) &= \sup_{Q \in \mathcal{O}_0: \text{TV}(Q, P_O) \leq \rho} \bar{\Gamma}_0(Q) + \Omega_L(\rho). \end{aligned}$$

Then

$$\mathcal{I}_1^\rho(P_O) \subseteq [L_\rho(P_O), U_\rho(P_O)] \cap [0, H_{P_O}(Y \mid T, C)].$$

Conditional-entropy continuity bounds can be obtained from [Audenaert \[2007\]](#), [Winter \[2016\]](#). The region is valid; matching sharp moduli require a separate perturbation analysis.

D Experimental Details and Additional Results

D.1 Simulation reproducibility

The simulation script is included at `scripts/simulate_finite.py`. It fixes the random seed at 20260901 and generates all simulation figures and tables in the manuscript. The outcome, proxy, and report alphabets are binary. Three latent states are used. Their extreme-coordinate distributions are

$$\begin{array}{lll} a_1 = (0.80, 0.20), & b_1 = (0.70, 0.30), & c_1 = (0.65, 0.35), \\ a_2 = (0.30, 0.70), & b_2 = (0.25, 0.75), & c_2 = (0.35, 0.65), \\ a_3 = (0.55, 0.45), & b_3 = (0.50, 0.50), & c_3 = (0.55, 0.45). \end{array}$$

These coordinates are mapped through the symmetric floor matrices associated with $(\alpha, \eta, \delta) = (0.05, 0.10, 0.10)$. The endpoint coverage experiment samples directly from the resulting eight-cell multinomial law.

D.2 NHANES construction and survey inference

The NHANES analysis joins DEMO, PAQ, minute-level PAXRAW, and 2019 public-use linked-mortality records by respondent sequence number. Artifact counts at the device maximum are replaced by the rounded mean of adjacent non-artifact minutes. Nonwear is defined by a 60-minute zero-count window permitting at most two short interruptions with counts no greater than 100. A day is valid with at least 600 wear minutes, and participants require four valid days. CPM is total valid-wear counts divided by total valid-wear minutes.

PAQ180 levels 1–4 define T . The CPM thresholds are weighted quantiles computed once on the primary complete-case sample: 295.806 for two bins; 242.523 and 360.826 for three bins; and 212.612, 295.806, and 397.614 for four bins. The primary table contains 6,565 participants, split as 3,317 from 2003–2004 and 3,248 from 2005–2006. The pooled examination weight is WTMEC2YR/2. Bootstrap draws resample PSUs within cycle-specific survey strata, and conditional permutations shuffle the proxy within (T, C, cycle) cells. The primary comparison is ten-year mortality with four CPM bins; the other five rows in [Table 5](#) are prespecified robustness analyses rather than a multiplicity-controlled family of confirmatory tests.

The positive trend across two, three, and four proxy bins is interpreted as representation sensitivity, not as a formal monotonicity claim. Finite-table plug-in mutual information can increase with alphabet size, so the paper reports the bias-corrected estimates and full-pipeline conditional permutations alongside the plug-in values. Cutpoints, processing rules, and the primary row were frozen before the outcome-linked result was inspected.

D.3 NHANES boundary compatibility and representation sensitivity

For the canonical $4M \times 4T \times 4C$ representation, only $(\alpha, \eta, \delta) = (0, 0, 0)$ is plug-in compatible. An ordinary bootstrap cannot represent sampling uncertainty for an empirically absent category because that category remains absent in every resample. We therefore also apply a boundary-safe outer screen. Within each context stratum, the weighted empirical law is written as a weighted sum of PSU-level probability vectors. A union bound over sign vectors gives a simultaneous ℓ_1 radius; the four context-specific radii range from 1.036 to 1.162. The resulting screen does not reject any of the 216 grid points, but it certifies none. This is an intentionally conservative, low-power conclusion: positive floors remain statistically unresolved and scientifically uncalibrated in the canonical table.

We additionally report post-canonical coarsening diagnostics without pseudocounts. These analyses change the representation or conditioning set and therefore change the estimand. Table 7 gives selected diagonal settings with $\alpha = \eta = \delta = \lambda$. The intervals are plug-in sharp sets for the stated coarsened law; the minimum-coordinate intervals are bootstrap stability diagnostics rather than a substitute for external floor calibration.

Table 7: Post-canonical NHANES representation sensitivity at a common floor λ . All rows use ten-year mortality and two CPM bins. “C1” pools context; “age2” conditions only on age group; “C4” is the primary sex-by-age context.

Representation	λ	min-coordinate 95% interval	sharp lower	sharp upper
C1, four-level T	0.0050	[0.00058, 0.00179]	0.03597	0.39583
C1, four-level T	0.0100	[0.00022, 0.00143]	0.03714	0.36045
age2, four-level T	0.0050	[-0.00019, 0.00035]	0.00997	0.29950
C4, two-level T	0.0050	[0.00021, 0.00203]	0.01280	0.30245
C4, two-level T	0.0100	[-0.00059, 0.00143]	0.01327	0.26707

The unadjusted C1 representation remains point compatible at $\lambda = 0.01$ in each survey cycle separately. This does not rescue the canonical conditional estimand: pooling C can reintroduce age and sex composition, and coarsening T or M changes both the witness and the scientific meaning of a probability floor. The table is therefore evidence that sparse-cell compatibility is representation-dependent, not evidence selecting one preferred positive restriction.

D.4 CrossCheck construction

The cleaned CrossCheck table aligns behavioral features with repeated EMAs [Adler et al., 2022]. We retain adjacent assessments separated by one to seven days and require the current row to contain at least three of four declared sensors: location distance, conversation duration, unlock duration, and mean light. The standardized sensor composite and current negative-symptom EMA are divided into participant-equal-weight tertiles; the same EMA thresholds define the next-assessment outcome. Median sensing quality defines two context strata. Each participant receives total weight one, so the independent unit for uncertainty is the participant rather than the assessment pair.

Table 8: CrossCheck longitudinal witness with participant-level inference and 1,000 bootstrap draws.

Participants	pairs	plug-in $\widehat{W}$	bias-corrected	95% CI	permutation p
60	5,607	0.01037	0.00121	[-0.01182, 0.01037]	0.02797

The interval crossing zero governs the manuscript conclusion. The permutation diagnostic and the cluster interval answer different finite-sample questions and are not combined into a significance claim.

D.5 NHANES and CrossCheck reproduction record

The formal protocol was frozen as `nhanes-crosscheck-formal-v1` before the outcome-linked NHANES result was inspected. It fixes seed 2026090501, the processing and inclusion rules, cutpoint construction, survey weights, resampling units, and 1,000 bootstrap and permutation replicates. A complete same-seed rerun reproduced all 15 prespecified core CSV files byte for byte. The

post-canonical positive-floor sensitivity analysis fixes seed 2026090502; its seven CSV outputs and the four outputs of the canonical boundary screen were also byte-identical in independent reruns. These checks establish computational repeatability of the declared pipelines. They do not turn a nonrejected channel floor into a scientifically calibrated restriction.

D.6 MELD data construction

The compact analysis file contains 2,330 adjacent-turn pairs from 268 MELD dialogues. Feature extraction was performed before the test analysis. Transcript TF-IDF and truncated SVD, audio PCA, and visual PCA were fitted on training data. The released compact table contains only the transformed vectors, sentiment labels, pair identifiers, and context variables. Its archive hash was checked before analysis.

The next-turn outcome counts are 750 negative, 1,117 neutral, and 463 positive. The context variables are the current-speaker major-character indicator and an indicator that the next turn is spoken by the same character. The primary analysis uses all 32 text dimensions, 16 audio dimensions, and 32 visual dimensions. The fused input concatenates the audio and visual vectors.

D.7 Repeated-split stability

Table 9: Cross-fitted log-score gains across eight dialogue split seeds under the canonical protocol. Values are in bits per next-turn pair.

Seed	Audio	Visual	Audio+visual
11	0.00149	-0.00165	0.00056
23	0.00287	0.00137	0.00280
37	0.00190	0.00133	0.00174
53	0.00069	0.00096	0.00097
71	0.00024	-0.00012	-0.00027
89	0.00059	-0.00056	0.00068
107	0.00117	0.00254	0.00483
131	0.00112	-0.00123	-0.00043

D.8 Same-speaker transitions

For the 565 transitions in which the current and next utterances share a speaker, the estimated gains are shown in [Table 10](#).

Table 10: Exploratory same-speaker next-turn analysis.

Modality	gain	95% lower	95% upper
Audio	0.00020	-0.01025	0.01064
Visual	0.00715	-0.00334	0.01765
Audio+visual	0.00883	0.00018	0.01749

The fused interval is slightly above zero under this refitted subgroup protocol. The analysis is exploratory and is not multiplicity-adjusted across subgroups or modalities.

D.9 Representation ladder

Table 11: Incremental log-score gain under nested text-SVD representations.

Text dimensions	Audio	Visual	Audio+visual
4	0.00250	-0.00166	0.00146
8	0.00268	-0.00277	0.00085
16	0.00274	-0.00214	0.00144
32	0.00293	-0.00125	0.00103

The canonical ladder does not show a monotone decline in non-report gain as the text representation is enriched. All corresponding confidence intervals include zero; we therefore use the ladder only as a representation-sensitivity diagnostic.

D.10 Exploratory general-report endpoint calculations

For the numerical endpoint check, the continuous 32-dimensional report representation is retained and each non-report representation is mapped to four train-fitted clusters. Dialogue-cross-fitted multinomial models estimate $P(Y, M \mid R, C)$. The common report mass is fixed at $\delta = 0$ because the compact data do not provide an externally fixed reference distribution ν . [Table 12](#) reports selected model-based functionals.

Table 12: Exploratory model-based sharp functionals for the continuous report representation. The lower endpoint uses $\eta = 0.02$ and the upper endpoint uses $\alpha = 0.05$.

Modality	model witness	lower endpoint	upper endpoint
Audio	0.00495	0.00585	0.92755
Visual	0.00228	0.00273	0.92889
Audio+visual	0.01189	0.01449	0.92615

The calculation demonstrates the population functional on a continuous report representation. Its values depend on the fitted conditional model and should be read alongside the dialogue-grouped predictive analysis in [Table 6](#).

D.11 Canonical MELD reproduction record

The released MELD computation freezes the two lightweight input-table hashes, the Python and package versions, the primary outer and inner fold assignments, tuning choices, random seeds, and multiplier configuration. The canonical environment records Python 3.14.5, NumPy 2.5.2, and scikit-learn 1.9.0. The archive verifier reports passing integrity, privacy, documentation, hygiene, and reproducibility gates. A duplicate complete run agrees with the canonical reference on all 111 compared numerical cells. The historical interactive analysis did not retain an equivalent fold and environment record; its numerical summaries are therefore used only for audit comparison and are not reported as manuscript results.